VADIM FEDOROV

vadimfedorov.sv@gmail.com | (408) 591-1286 | Sunnyvale, CA
linkedin.com/in/vadim-fedorov-sv | github.com/vadimfedorovsv

EDUCATION

University of California, Davis Expected June 2028
Bachelor of Science, Electrical Engineering *Davis, CA*
GPA: 3.7 • Dean's List: Fall 2025, Winter 2025 • Neurotech Projects Division

TECHNICAL SKILLS

Programming & Firmware: Python, C/C++, embedded C, Java, MATLAB, Arduino, Git

Hardware & Electronics: PCB design, circuit design, I²C, Teensy, sensor integration, soldering, prototyping

Robotics & Neurotech: Teleoperation, micro-ROS, PyBullet, robotic actuation, EEG signal processing, brain–computer interfaces, 3D-printed prosthetics

Tools & Software: SolidWorks, KiCad, Linux, OpenBCI

Languages: English, Russian

PROJECTS

Krysalis Hand Jan 2026 – Present
Embedded Systems & Teleoperation Lead

- A 19-actuator robotic hand built for real-time dexterous teleoperation: a Manus motion-capture glove maps an operator's hand directly onto self-locking finger and thumb joints, geared finger abduction/adduction, and wrist rotation through a PyBullet and micro-ROS pipeline.

BCI Prosthetic Leg Dec 2025 – Jun 2026

- Engineered a 3D-printed prosthetic leg driven by EEG-based brain–computer interface control — decoding neural signals into coordinated knee and ankle articulation for assistive mobility.

PyBCI App Jan 2026 – Present

- Built a unified EEG research platform — the first to consolidate recording, preprocessing, paradigm presentation, model training, and visualization into a single environment, replacing the fragmented stack of separate tools the field currently relies on.

EXPERIENCE

Laboratory for AI, Robotics, and Automation (LARA) Jan 2026 – Present
Undergraduate Robotics Researcher | Davis, CA

- Restored a non-functional teleoperation pipeline for Krysalis, a 19-DOF robotic hand that crashed after ~2 seconds of operation, by independently diagnosing I²C bus contention on the receiving microcontroller and rewriting firmware on both inbound and outbound signal paths — achieving stable, indefinite real-time control at ~50 ms end-to-end latency.
- Redesigned the system's custom control PCBs to harden I²C communication and power delivery — adding signal pull-up resistors and dedicated voltage regulation to protect the onboard Teensy microcontroller from power-related failure.
- Extended teleoperation from the hand to wrist articulation by integrating two base-mounted rotary actuators, advancing the platform toward full-arm dexterous manipulation.

Cherry Chase Pet Hospital Jun 2024 – Sep 2025
Customer Service Representative | Sunnyvale, CA

- Managed high-volume client communications across phone, email, and front-desk channels, maintaining a professional experience during peak clinic hours.
- Coordinated appointment scheduling, client intake, and administrative workflows, streamlining front-desk operations in support of veterinary staff.

PUBLICATIONS

The Implementation of Ethical AI Within Autonomous Vehicles Sep 2023 – Feb 2024
Researcher & Writer

- Published in the Journal of Student Research; analyzes ethical decision-making frameworks for autonomous vehicles. View paper